Settlement Advancement & Regional Economy

MY MEDIEVAL LIFE | STANDALONE EXPANSION DESIGN SPECIFICATION

Status: owner-directed concept approved in principle • Implementation build: to be scheduled after current stabilization • Companion: Mining & Industrial Development Expansion

1. Executive design decision

Separate the progression of villages into towns and major settlements, and the regional trade system that results, from the dedicated Mining & Industrial Development Expansion. Mining develops what an industry can do; this expansion develops what a settlement collectively knows, produces, needs, trades, and becomes.

Permanent design rule: a caravan is not an automatic daily income source. The quantity of money and goods arriving depends on whether another settlement needs what is offered, can afford it, and finds the trade worthwhile. A settlement with no competitive or useful export may receive little or no outside money.

For Builds 15–16, the current 30-coin daily caravan allowance may remain an explicitly labeled, adjustable temporary simulation stand-in. It is not an approved permanent economic rule. Do not disrupt ongoing equilibrium validation to implement this expansion.

2. Two independent but connected expansions

Expansion	Primary responsibility	Examples
Mining & Industrial Development	Expand mining-related careers, recipes, equipment and extraction techniques	Surface ore, prospectors, assayers, quarries, open pits, underground workings, carts and track
Settlement Advancement & Regional Economy (this document)	Advance settlement capabilities and link their supply/demand to other settlements	Village skills, town growth, specializations, regional inventories, trade demand, merchant budgets, competition

Dependency: mining levels provide one example of settlement capability. The settlement progression framework must be generic enough to support farming, forestry, masonry, smithing, trade and future industries without requiring all of them to be implemented at once.

3. Settlement growth is emergent, not an upgrade button

Settlements may grow from hamlets/villages to towns and major settlements through sustained population, homes, civic capacity, economic output, services and trade connections. These labels describe development, not universal tech tiers. Growth is not guaranteed or permanent: shortages, migration, disasters, poor governance and failed industries can cause stagnation or decline.

- No automatic technology unlock merely because a population threshold is crossed.
- No universal ladder where every town has the same industries. A small mining village can outclass a larger farming town in mining.

- Buildings, tools, trained residents, accessible resources and knowledge transmission must underpin real capabilities.
- Villagers remain autonomous. The player may accelerate, disrupt, fund or exploit growth, but must not be necessary for it.

4. Industry capability levels belong to settlements

Each settlement maintains an industry capability profile, derived from the actual capabilities of its residents and infrastructure. Individual NPC skills remain distinct; settlement proficiency does not magically grant expertise to every villager.

Mining example	Settlement capability	Required real-world support
Level 1	Surface deposit extraction	Exposed ore, basic miners, picks, hauling
Level 2	Prospecting and assaying	Trained prospectors/assayers, forged kits, geological clues, sample testing
Level 3	Open-pit excavation and quarrying	Earth removal, stone processing, ramps, hauling, carpentry and forge capacity
Level 4	Underground excavation	Tunnel engineering, supports, tracks, minecarts, underground navigation and safety

These are approved mining progression concepts; specific thresholds and cross-industry prerequisites are to be designed and tested, not assumed fixed. An advanced mining operation may depend on capable smiths, carpenters, haulers and builders.

Capability progression mechanics

- Experience: successful work, practice, apprenticeships and problem-solving produce knowledge and competence.
- Specialists: qualified NPCs enable advanced activities; recruiting or training them takes time and resources.
- Infrastructure: appropriate workshops, storage, transport, tools and construction must exist and remain functional.
- Knowledge continuity: instruction, apprenticeship and records preserve skills across generations; population loss can erode capacity.
- Industry interdependence: advanced operations require actual outputs of supporting professions, not abstract research points.
- Settlement specialization: local geology, land, labor, demand and investment encourage distinctive economies.

5. Regional economy replaces the caravan quota

A caravan represents an originating settlement or trade network. Ultimately it carries actual goods and finite purchasing power based on external inventories, needs and means. It must not buy every available product by default.

A viable trade transaction requires

- Demand: a destination settlement has a meaningful unmet need or profitable resale opportunity.
- Supply: the source settlement has real goods available after its own competing uses and commitments.
- Purchasing power: the buyer has money or acceptable barter/credit terms; need alone does not create wealth.
- Price and competition: total cost, quality, distance, risks and alternative suppliers make the trade worthwhile.
- Logistics: cart capacity, roads, travel time, losses, security and storage constrain quantity.
- Accounting: physical inventory and currency move between actual owners. No hidden duplication or infinite cash.

6. Illustrative economic interactions

Settlement A	Settlement B	Outcome
Agricultural village: grain surplus	Mining town: food deficit and money	Merchant buys grain if price and transport work
Mining town: ore surplus	Farming village: needs metal tools	Ore or tools flow outward; food may flow back
Timber village: lumber surplus	Masonry town: no lumber demand	No guaranteed sale despite surplus
Starving town: high food need, low cash	Food-producing town: surplus	Trade may fail without barter, credit or another income source
Two towns selling identical goods	Third town seeking supplies	Competition, price and distance influence which exporter earns the sale

Demand is not an automatic entitlement to imports. A settlement can need goods it cannot afford, and another settlement can produce goods that no one wants.

7. Caravans: transition from placeholder to simulated merchant

Current phase: Builds 15–16

- Retain the provisional 30-coin/day mechanism only as needed for baseline stability; label it clearly as an abstract external-market budget.
- Measure net coin inflow/outflow, import/export quantities, inventory conservation and dependence on imported ore/coal.
- Keep the budget configurable and expose its effects in simulation reports; do not silently treat it as real demand.

Intermediate phase: regional-market abstraction

- Create abstract off-map settlements with distinct finite stocks, needs, productive strengths and budgets.
- Generate caravan orders from modeled demand and purchasing power rather than a fixed daily purchasing obligation.

- Use explicit restocking and settlement-accounting rules; avoid unexplained daily currency creation.
- Support variable arrivals and empty trips when no worthwhile trade exists.

Mature phase: fully simulated regional settlements

- Trade routes link actual settlements with persistent populations, production, consumption, inventories and treasuries.
- Merchants select profitable or necessary routes subject to competition and transport capacity.
- Regional supply disruptions and new industrial skills change prices, routes and settlement fortunes.
- The player can participate in or disrupt these relationships without becoming the central economic engine.

8. Settlement growth systems and their interlocks

- Population and housing: births, partnerships, households, migration, jobs, construction materials and capacity must agree.
- Industry: training and specialist availability determine what can be made and exported.
- Infrastructure: roads, bridges, warehouses, marketplaces, workshops and carts affect costs and throughput.
- Civic systems: leadership, budgets, public works and security may enable or hinder growth, using finite funds.
- Ecology and land: farming, forests, mines, quarries and wildlife remain finite and affected by local activity.
- Knowledge and generations: experience can be taught, recorded or lost; no omniscient settlement skill bonus.
- Reputation and diplomacy: reliability, conflict and trade history may affect partners, subject to knowledge provenance.

9. Failure modes and economic safeguards

- Do not equate settlement tier with every industry level; avoid uniform cookie-cutter cities.
- Do not award skills solely by time elapsed or abstract points disconnected from work.
- Do not grant free imports, money, ore, food, tools or building materials to keep an economy looking healthy.
- Avoid permanent guaranteed export demand; settlements can lose customers and decline.
- Prevent circular money printing via merchant budgets, resale, taxes or trade accounting.
- Allow local setbacks without declaring the entire world failed; test adaptation and long-term survival.
- Keep mining deposits finite. Continued mining employment depends on discovery, new workings, trade and real labor.

10. Suggested implementation phases (not assigned build numbers)

Phase	Scope	Release evidence
A — Design and ledger	Requirements IDs, cross-industry skill schema, save compatibility, dependency audit	Validated register, design docs, explicit assumptions

B — Settlement capability model	Per-industry capabilities, NPC knowledge, workshops, prerequisites, decay/continuity	Unit tests for unlocks, loss of specialists, generational continuity
C — Abstract regional economy	Off-map settlements, finite demand/supply/budgets, variable caravan trade	Accounting and 100/365-day multi-seed tests
D — Settlement development	Growth/decline, specialization, housing, infrastructure, population and employment	Multi-generation simulations across settlement types
E — Actual intersettlement network	Persistent multiple settlements, routes, logistics, competition, merchant AI	No duplicated inventory/currency, trade route and disruption tests

The mining expansion may proceed as a separate build, tentatively Build 17 or 18; the schedule for this settlement/regional economy expansion is intentionally unassigned. Avoid scope creep into Build 15–16.

11. Acceptance criteria and simulation gates

- A village can prosper or decline without direct player intervention, with causes traceable to real inputs.
- At least two settlements can develop different industry profiles despite different population sizes.
- A settlement cannot perform advanced work without appropriate people, knowledge, tools and infrastructure.
- A caravan may arrive with substantial buying power when real demand and funds exist, or buy nothing when demand is absent.
- Imported and exported goods and money reconcile exactly across merchant and settlement inventories.
- A starving but cash-poor town does not conjure purchasing power; trade failure is observable.
- No fixed 30-coin quota remains in the final demand-driven implementation.
- Multi-seed 100-day, 365-day and multi-generation tests examine production, food, housing, population, industry employment, prices, trade and resilience.
- Existing save games and established gameplay systems continue to work; migration is documented.

12. Decisions settled and questions deferred

Approved direction

- Mining/industrial extraction is its own expansion, not an emergency patch to Builds 15–16.
- Settlement advancement and regional economy is a separate, connected expansion.
- Settlement capabilities are industry-specific and earned through people, skills and infrastructure.
- Caravan buying power ultimately comes from other settlements' actual needs and means, not a permanent allowance.
- Current 30-coin daily mechanism is a temporary placeholder only.

To be designed during expansion planning

- Exact settlement classifications and thresholds, if any.

- Detailed skill progression rates, knowledge transfer, decay and specialist certification.
- Off-map market model and how money/production are accounted for before full map expansion.
- Trade route scheduling, credit/barter rules, taxation and merchant ownership.
- Which cross-industry prerequisites are mandatory versus optional improvements.

13. Claude handoff and change-control directive

Register this expansion as a distinct major initiative in the permanent requirements ledger and Interlocking Systems Manifest. Cross-reference, but do not merge it into, the Mining & Industrial Development Expansion. Preserve the owner's decisions verbatim in decision records. First audit existing economy, professions, caravan code, settlement representation, persistence and simulation harness. Then propose phased implementation with testable requirements, dependencies and migration plans. Continue currently approved Build 15–16 stabilization and simulations without introducing this large feature prematurely.

Design principle: a settlement becomes economically important because of what its people can sustainably produce and what other settlements genuinely need—not because a game system grants it a daily customer.